

1. Consider a two-mass- two-spring system arranged in a linear chain. Masses p_1 and p_2 are connected by two ideal springs with spring constants k_1 and k_2 as follows:

rigid wall $\rightarrow$ spring $k_1 \rightarrow p_1 \rightarrow$ spring $k_2 \rightarrow p_2$.

Assume the motion is one-dimensional and frictionless.

Write down the Lagrangian of the system in terms of the generalized coordinates.

[2]

2. Derive the characteristic (eigenvalue) equation for the normal modes. Hence, calculate the normal-mode (eigen) frequencies. Take $p_1 = p_2 = m, k_1 = k_2 = k$

[3]

known $M\ddot{x} = -Kx$ for Spring Mass System.

3. Find the eigenvectors corresponding to the normal modes and interpret them as normal-mode shapes.
Take $p_1 = p_2 = m, k_1 = k_2 = k$

[3]

$$(k - m\omega^2) \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = 0$$

4. If both masses are released from nonidentical initial conditions, briefly describe the qualitative nature of their subsequent motion. Take $p_1 = p_2 = m$, $k_1 = k_2 = k$ [1]

Both masses will oscillate with the same frequency (phase diff = 0,

5. Provide pseudocode (Euler method) to integrate the differential equation $\dot{x} = a \cdot x - b$, with parameters $a = 1$; $b = 2$, and initial condition $x(t = 0) = 1$. Analytical solution not required. [2]

$$\frac{dx}{dt} = ax - b$$

6. Explain briefly how the linear system $Ax = b$ can be solved using LU decomposition? (No pseudocode required.) [2]

7. Write pseudocode/algorithm for solving $Ax = b$ when A is lower-triangular matrix assuming a unique solution. [2]

$A \rightarrow$ lower triangular (upper than diagonal elements are zero)



8. Consider the discrete map $x_{n+1} = 2 * x_n(1 - x_n)$, where $n = 0, 1, 2, \dots$, and $x_0 = 0.3$. Explain what happens to x_n as n becomes very large? [2]